

Reading Support: Linguistics as a Special Science

Companion to Week 4

Cem Bozşahin

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Mathematical thinking does not determine the science

Supports: Mathematical thinking does not make linguistics a branch of mathematics

The use of mathematics does not make linguistics a branch of mathematics, any more than mathematical physics becomes pure mathematics. The empirical science determines which objects, measurements, and correspondences require explanation.

Mathematical languages also contain presuppositions about sets, rules, countability, and identity. Formalization helps expose them when the route from linguistic observation to formal object remains explicit.

MOLR §2.1

Facts become scientific evidence through comparison

Supports: The scientific object is produced through disciplined comparison

Science does not merely collect singular facts. It seeks relations that remain invariant across selected facts. Selection requires a conceptual vocabulary, and that vocabulary determines which similarities and differences become visible.

The practice remains corrigible because others can reconstruct the abstraction, examine its consequences, and compare the proposed invariant with further variants.

MOLR §2.1

Children's innovations reveal productive organization

Supports: Children do more than repeat adult forms

Utterances such as “It colds my bottom” and “Mummy trousers me” need not have been copied from adult speech. They show a child extending familiar structural possibilities to lexical material that adult convention does not ordinarily admit.

Children rise above the heard forms by relating sentences to situations and activities. Their errors reveal what has been generalized.

MOLR §2.2

Linguistic competence includes practical mastery

Supports: Knowing that presupposes knowing how

Speakers can reliably produce, interpret, and repair expressions without being able to formulate the rules attributed to them. Knowing that a rule holds presupposes practical criteria for applying it to cases.

An explicit grammar describes part of this organization. It cannot replace the capacity whose regularities supplied the evidence for the description.

MOLR §2.2

Acceptability reflects interacting conditions

Supports: A star compresses heterogeneous judgments

The contrast between “win for an hour” and “play for an hour” depends on event structure and the temporal modifier. Alternative contexts can sometimes repair a starred form by changing the understood event.

Judgments therefore provide evidence about the interaction of grammatical, lexical, semantic, and contextual admissibility. The analysis must identify which condition produces the contrast.

MOLR §2.2

Ontogeny, continuity, and diversity constrain one another

Supports: Three questions constrain any theory of language

An account of acquisition must explain how children arrive at a system from available linguistic activity. A continuity account must relate developing systems to adult grammars without assuming either identity or an unexplained rupture.

The diversity problem asks why languages realize apparently limited dependency types in so many different ways. A proposal that answers one question by making another impossible has not solved the combined problem.

MOLR §2.3

Open structural capacity differs from actual sentence length

Supports: Long-distance dependency is finite in every token, open in the system

Every observed dependency crosses finitely much material. The grammar may nevertheless admit the relevant construction inside another construction of the same kind, so no fixed bound belongs to the description.

Processing limits and communicative choices sharply constrain actual performance. They do not by themselves show that the structural relation has a maximum grammatical depth.

MOLR §2.3

Mathematicisation includes the bridge to the phenomenon

Supports: Using mathematics and mathematicising the object are different

Applying a mathematical technique to previously selected data uses mathematics in linguistics. Mathematicising linguistics also asks how linguistic objects became the arguments and values of the formal system.

The stronger task includes the admissibility conditions, the structure-preserving correspondences, and the empirical grounds for treating the mathematical result as a result about language.

MOLR §2.4